

LiveLocalizer: Augmenting Mobile Speech-to-Text with Microphone Arrays, Optimized Localization and Beamforming

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ABSTRACT

Speech-to-text capabilities on mobile devices have proven helpful for language translation, note-taking, hearing and speech accessibility, and meeting transcripts. However, their usefulness is constrained by being unable to distinguish between multiple speakers, track which direction speech is coming from, and provide acceptable performance in noisy environments.

This work introduces efficient real-time audio localization and adaptive beamforming algorithms on custom sound perception hardware running on a low-power microcontroller and four integrated microphones. A prototype is implemented in a phone case form factor and is plug-and-play with modern smartphones.

We characterize the performance in technical evaluations of localization, beamforming, and diarization. We demonstrate how the phone case extends existing smartphones with speaker diarization in a speech-to-text app, sound direction visualization, and sound enhancement through beamforming. In the future, we hope our approach will inspire the widespread adoption of advanced microphone arrays that natively unlock the potential of spatial sound processing and perception in mobile and wearable devices.

CCS CONCEPTS

• **Human-centered computing** → **Graphics input devices**; *Ubiquitous and mobile computing systems and tools*.

KEYWORDS

Speech-to-text, ASR, STT, audio, speech, microphone array, beamforming, accessibility

ACM Reference Format:

Artem Dementyev, Dimitri Kavensky, Samuel J. Yang, Mathieu Parvaix, Chiong Lai, and Alex Olwal. 2023. LiveLocalizer: Augmenting Mobile Text-to-Speech with Microphone Arrays, Optimized Localization and Beamforming. In *The 36th Annual ACM Symposium on User Interface Software and*

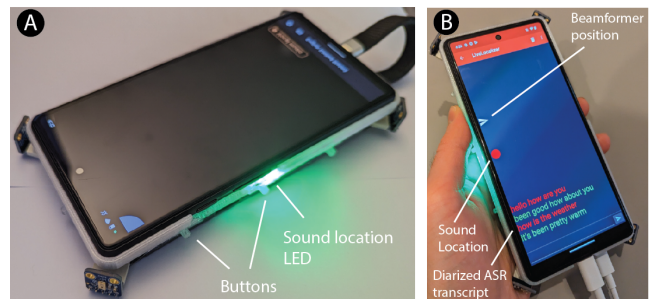


Figure 1: We implemented two interfaces to enable spatial speech perception using microphone array processing algorithms: A) a physical interface consisting of a custom phone case where LEDs indicate the direction of a localized sound source and buttons to select beamforming direction; and B) Mobile phone application interface with the same functionality integrated with a speech-to-text application with additional speaker diarization capabilities.

Technology (UIST '23 Adjunct), October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3615789>

1 INTRODUCTION

Recent advances have enabled real-time automatic speech recognition (ASR) in mobile and embedded hardware, supporting a range of conversational scenarios [1, 12]. Real-time ASR can enhance human communication in a variety of ways; real-time translation between languages, transcriptions during an interview, automatic subtitle generation, hearing accessibility [9, 14], and note-taking. However, there are still limitations with real-time speech-to-text — specifically, the ability to distinguish multiple speakers (*speaker diarization*), tracking the direction of speech (*localization*), and enabling robust performance in noisy environments.

Many of the challenges in the scenario stem from the spatial complexities of audio, which can not be perceived by a single-microphone ASR system. Many modern phones have three microphones, but their position does not allow precise localization. The benefits of multi-microphone topologies for localization have been

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UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

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ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3615789>

demonstrated in numerous applications, such as public safety [15], virtual reality [10], robot navigation [11], mobile computing [7, 13] and audio accessibility [5, 6].

Recent increases in computational power allow multi-microphone techniques to be used in low-power devices with low latency. In this work, we developed low-latency localization and beamforming algorithms that can run on general-purpose microcontrollers. For 360-degree localization and adaptive beamforming, we developed *LiveLocalizer*, a 4-microphone phone case with a stand-alone general-purpose low-power microcontroller. While we focus on mobile ASR, our hardware and software could also be generalized to other form factors and scenarios.

2 IMPLEMENTATION

Hardware. We designed the sound perception system based on a 110 MHz M33 ARM Core (STM32L55, STMicroelectronics), four digital pulse density modulation (PDM) microphones (MP34DT01-M, STMicroelectronics), and an audio codec chip (SGTL5000, NXP) routed to a 3.5 mm audio jack. See Figure 2A. We mounted the microphones in the four corners of a custom-designed printed circuit board (PCB) to maximize spatial separation, as shown in Figure 2B. The PCB was screwed to a custom-designed phone case, 3D-printed in clear PETG material. We included interface elements for quick access and feedback; eight tactile buttons were placed on the edges of the PCB, and twelve RGB LEDs (WS2812b, Neopixel) were positioned on the periphery of the phone case. The device includes two interface options: USB-C for data and audio transfer and analog audio input and output. Also, we developed a compact prototype without the LEDs and buttons, as shown in Figure 2C. This device has a smaller main PCB and a flexible PCB for microphone routing.

Localization. For localization, the delay between all unique pairs of microphones needs to be estimated. We implemented a variant of Generalized Cross-Correlation with Phase Transform (GCC-PHAT) [8], which takes advantage of Fast Fourier Transform (FFT) [3] to be more efficient than standard, computationally expensive cross-correlation. Considering the uncertainties of sound propagation, a statistical approach using kernel density estimation (KDE) has to be used to determine the actual source location.

Beamforming. Beamformer is a spatial filter where sound delays are used to boost audio arriving from certain directions. We use a Filter and Sum (FAS) beamforming approach [4], which is needed for broadband speech signals. In our application of FAS beamforming, the beam can be steered to any azimuth angle (θ) by adjusting the steering weights, which are computed from the phase delays of the microphones.

Technical Evaluation. To quantify azimuthal localization ability, we mounted the device on a rotating platform with a stationary speaker playing either speech or noise audio. The results show that sound could be localized reliably for both speech and noise, with different loudness, and with 360° coverage. In normal conversational loudness (60–65 dB), the error was 11.1 to 22.1 degrees. The measured latency to reliably determine sound direction ranged from 70–500 ms, depending on speech characteristics. The power consumption of the whole system was 28 mAh.

To evaluate diarization accuracy, we set up four speech sources to mimic a turn-taking conversation with four talkers. We saw a

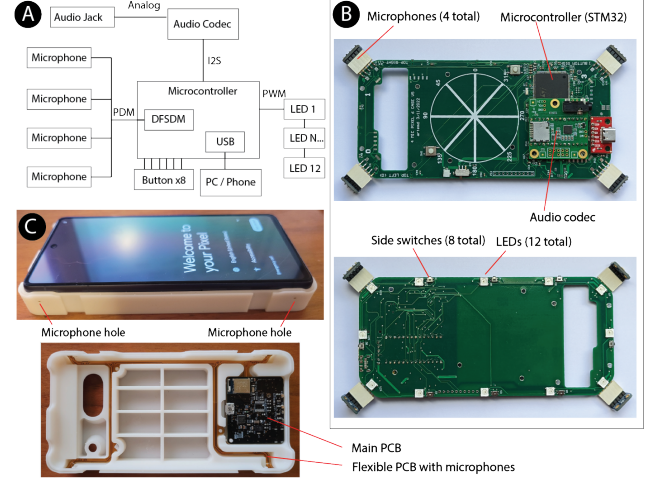


Figure 2: Overview of the LiveLocalizer phone-case device. A) System diagram illustrating how audio input from the four microphones is processed. B) Printed circuit board with top view on the left and bottom view on the right. C) Compact prototype with miniaturized microphones and PCB. Routing to microphones is replaced by a flexible PCB, as seen when the phone is removed in the bottom image.

consistent diarization error rate (DER) improvement of 23 to 35 % for four microphone configurations in comparison to a three-microphone setup.

3 END-TO-END SPATIAL SPEECH-TO-TEXT APPLICATION.

LiveLocalizer could be used in two ways. First, by physically interacting with the phone case where LEDs light up to indicate the direction of the sound source, as seen in Figure 1A. Additionally, the side buttons can be used to move the beamformer in the direction of the pressed button. With this standalone UI, the device can be attached to and used with the phone without communicating with a phone application. Second, *LiveLocalizer* could be integrated with a mobile phone application (Figure 1B). To showcase this, we implemented a mobile ASR application on Android that uses localization and beamforming functionality. The application has the following features:

Sound source visualization. We use a circle overlay on the edge of the screen to indicate sound direction. The circle moves according to the azimuth angle of the audio, as can be seen in Figure 1B. Alternatively, a vertical line can be used if only 180-degree localization is available, such as from the two built-in phone microphones. We change the size of the circle and line according to the localization confidence, corresponding to the peak height in the azimuth probability estimate, with a larger circle or line denoting more confidence.

Speaker diarization in transcription We use the Android Speech Recognition API [2] to obtain real-time transcription and display the results on the mobile phone screen as vertically scrolling captions. The transcript is colored by directly mapping the sound

arrival angle to the 360° color wheel. In such mapping, the text is colored green if sound arrives from the top of the phone (around 90°), and red for bottom arrival (270°).

Beamforming to a selected source. The microphone array can be used to beamform towards an audio signal to increase the gain in a specific direction and reduce the gain from audio in other directions. We connected the 3.5 mm jack audio output of LiveLocalizer to a mobile phone, where the beamformer could be controlled by dragging a white circle on the edges of the screen. The beamformer steering is recalculated as the white circle is moved.

4 CONCLUSION

This work demonstrates how microphone array signal processing can supplement speech-to-text by measuring and visualizing the spatial dimension of audio. To showcase the spatial dimension, we developed a custom hardware solution in the form factor of a phone case containing four microphones. We developed low-latency 360-degree localization and beamforming algorithms that can run on general-purpose low-power microcontrollers.

We developed two hardware interface options. Standalone buttons and LEDs on the phone case allow the device to function untethered to the phone or phone application. To showcase more complex interactions, we enable communication of audio and localization data to the ASR application on the phone. This integration allows for new capabilities available to the mobile phone application, including sound source location, speaker diarization, and user control of the beamformer, allowing for audio enhancement in a specific direction.

With this work, we demonstrate that microphone array processing can be integrated into new and existing low-power mobile devices, thereby leveraging the natural spatial properties of audio to enhance audio experiences.

ACKNOWLEDGMENTS

We thank Richard Lyon, Pascal Getreuer, Alex Huang, Sagar Savla, Dmitrii Votintsev, and the Google Interaction Lab.

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